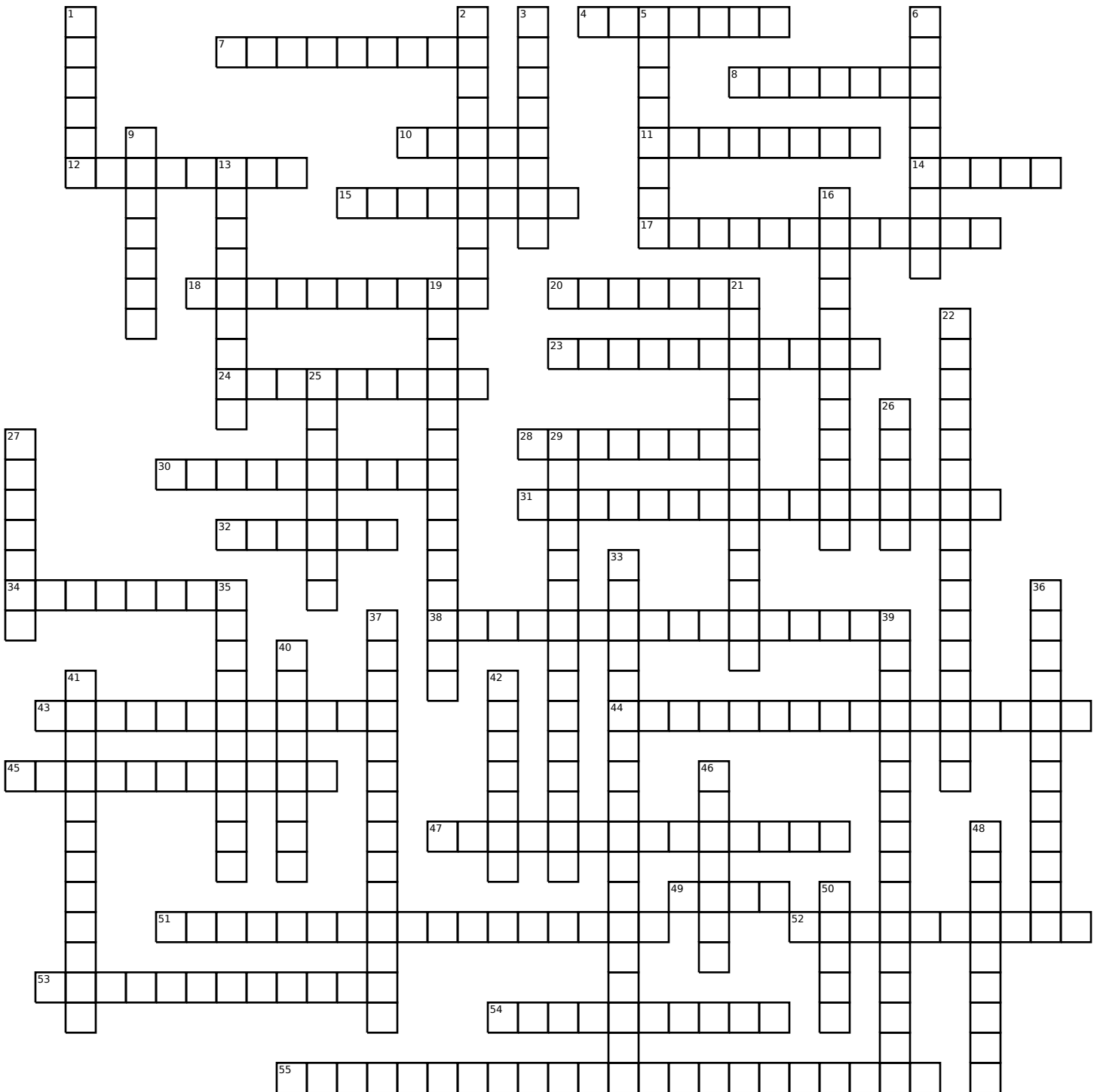


CH 27 terms (Campbell)

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Across

- 4 In many prokaryotes, a dense and well-defined layer of polysaccharide or protein that surrounds the cell wall and is sticky, protecting the cell and enabling it to adhere to substrates or other cells; or the sporangium of a bryophyte (moss, liverwort, or hornwort) .
- 7 A thick-coated, resistant cell produced by some bacterial cells when they are exposed to harsh conditions .
- 8 A short, hairlike appendage of a prokaryotic cell that helps it adhere to the substrate or to other cells .
- 10 A group of species that includes an ancestral species and all of its descendants. A clade is equivalent to a monophyletic group .
- 11 The smaller participant in a symbiotic relationship, living in or on the host .
- 12 A bacterial plasmid carrying genes that confer resistance to certain antibiotics . (2 Words)
- 14 An oriented movement toward or away from a stimulus .
- 15 A toxic protein that is secreted by a prokaryote or other pathogen and that produces specific symptoms, even if the pathogen is no longer present .
- 17 An organism that lives in environmental conditions so extreme that few other species can survive there. Extremophiles include extreme halophiles ("salt lovers") and extreme thermophiles ("heat lovers") .
- 18 In a specified group of organisms, a taxon whose evolutionary lineage diverged early in the history of the group . (2 Words)
- 20 A species or group of species whose evolutionary relationships are being examined in a given analysis .
- 23 The representation on a phylogenetic tree of the divergence of two or more taxa from a common ancestor. A branch point is usually shown as a dichotomy in which a branch representing the ancestral lineage splits (at the branch point) into two branches, one for each of the two descendant lineages . (2 Words)
- 24 An ecological relationship between organisms of two different species that live together in direct and intimate contact .
- 28 The plasmid form of the F factor . (2 Words)
- 30 A specialized cell that engages in nitrogen fixation in some filamentous cyanobacteria; also called a heterocyte .
- 31 Homologous genes that are found in different species because of speciation . (2 Words)
- 32 In Linnaean classification, the taxonomic category above genus .
- 34 A species or group of species from an evolutionary lineage that is known to have diverged before the lineage that contains the group of species being studied. An outgroup is selected so that its members are closely related to the group of species being studied, but not as closely related as any study-group members are to each other .
- 38 An organism that carries out only fermentation or anaerobic respiration. Such organisms cannot use oxygen and in fact may be poisoned by it . (2 Words)
- 43 Describing the group of bacteria that have a cell wall that is structurally less complex and contains more peptidoglycan than the cell wall of gram-negative bacteria . (2 Words)
- 44 A principle that states that when considering multiple explanations for an observation, one should first investigate the simplest explanation that is consistent with the facts . (2 Words)
- 45 In prokaryotes, the direct transfer of DNA between two cells that are temporarily joined. When the two cells are members of different species, conjugation results in horizontal gene transfer; or in ciliates, a sexual process in which two cells exchange haploid micronuclei but do not reproduce .
- 47 The use of organisms to detoxify and restore polluted and degraded ecosystems .
- 49 The larger participant in a symbiotic relationship, often providing a home and food source for the smaller symbiont .
- 51 As applied to DNA sequence data, a principle that states that when considering multiple phylogenetic hypotheses, one should take into account the hypothesis that reflects the most likely sequence of evolutionary events, given certain rules about how DNA changes over time . (2 Words)
- 52 An organism that absorbs nutrients from nonliving organic material such as corpses, fallen plant material, and the wastes of living organisms and converts them to inorganic forms; a detritivore .
- 53 Pertaining to a group of taxa that consists of a common ancestor and all of its descendants. A monophyletic taxon is equivalent to a clade .
- 54 An approach to systematics in which organisms are placed into groups called clades based primarily on common descent .
- 55 The transfer of genes from one genome to another through mechanisms such as transposable elements, plasmid exchange, viral activity, and perhaps fusions of different organisms . (3 Words)

Down

- 1 A technique for editing genes in living cells, involving a bacterial protein called Cas9 associated with a guide RNA complementary to a gene sequence of interest .
- 2 An organism that produces methane as a waste product of the way it obtains energy. All known methanogens are in domain Archaea .
- 3 A non-membraneenclosed region in a prokaryotic cell where its chromosome is located .
- 5 An organism that feeds on the cell contents, tissues, or body fluids of another species (the host) while in or on the host organism. Parasites harm but usually do not kill their host .
- 6 A staining method that distinguishes between two different kinds of bacterial cell walls; may be used to help determine medical response to an infection . (2 Words)
- 9 A small, circular, doublestranded DNA molecule that carries accessory genes separate from those of a bacterial chromosome; in DNA cloning, plasmids are used as vectors carrying up to about 10,000 base pairs (10 kb) of DNA. Plasmids are also found in some eukaryotes, such as yeasts .
- 13 ecological interaction that benefits each of the interacting species .
- 16 ecological interaction in which one organism benefits but the other is neither helped nor harmed .
- 19 An organism that requires oxygen for cellular respiration and cannot live without it . (2 Words)
- 21 A type of polymer in bacterial cell walls consisting of modified sugars cross-linked by short polypeptides .
- 22 The conversion of atmospheric nitrogen (N₂) to ammonia (NH₃) . (2 Words)
- 25 A common term for the two-part, latinized format for naming a species, consisting of the genus and specific epithet; also called a binomen .
- 26 In Linnaean classification, the taxonomic category above the level of family .
- 27 A taxonomic category, the second broadest after domain .
- 29 Homologous genes that are found in the same genome as a result of gene duplication . (2 Words)
- 33 An organism that thrives in hot environments (often 60–80°C or hotter) . (2 Words)
- 35 ecological interaction in which one organism, the parasite, benefits by feeding upon another organism, the host, which is harmed; some parasites live within the host (feeding on its tissues), while others feed on the host's external surface .
- 36 Describing the group of bacteria that have a cell wall that is structurally more complex and contains less peptidoglycan than the cell wall of gram-positive bacteria . (2 Words)
- 37 A method for estimating the time required for a given amount of evolutionary change, based on the observation that some regions of genomes evolve at constant rates . (2 Words)
- 39 An organism that lives in a highly saline environment, such as the Great Salt Lake or the Dead Sea . (2 Words)
- 40 An organism or virus that causes disease .
- 41 A process in which phages (viruses) carry bacterial DNA from one bacterial cell to another. When these two cells are members of different species, transduction results in horizontal gene transfer.
- 42 In bacteria, the DNA segment that confers the ability to form pili for conjugation and associated functions required for the transfer of DNA from donor to recipient. The F factor may exist as a plasmid or be integrated into the bacterial chromosome . (2 Words)
- 46 Similarity between two species that is due to convergent evolution rather than to descent from a common ancestor with the same trait .
- 48 A toxic component of the outer membrane of certain gram-negative bacteria that is released only when the bacteria die .
- 50 A taxonomic category above the species level, designated by the first word of a species' two-part scientific name .

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